ATHENS UNIVERSITY OF ECONOMICS AND BUSINESS
DEPARTMENT OF INFORMATICS

Wireless Networks and Mobile Communications
Fall Semester 2025-26

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Assignment #2 – 9 November 2025
Due date: 30 November 2025

The aim of the first experimental assignment is to get familiar with and execute experiments with the speed of a wireless interface and the throughput at the transport layer with the TCP and UPD protocols.

Devices that you will need: i) a laptop or desktop computer with Wi-Fi connectivity, ii) a smartphone, and iii) a wireless Access Point, which can be within the DSL router of your broadband connection (referred to below as “AP”). Instead of a smartphone you can use a second laptop or desktop computer and the opposite: instead of a laptop you can use a second smartphone. When conducting the measurements you are advised to keep your smartphone and laptop on a stable surface.

Software that you will need: i) the application iPerf 3.0 for measuring the throughput at a smartphone and laptop/desktop, which is is available at https://iperf.fr/ and ii) a Wi-Fi analyzer application for a laptop/desktop.

The report you submit must contain the topology where you conducted the measurements, the device type (smartphone or laptop/desktop), and the IP addresses of the devices. The report must include the tables with your measurements, the graphs that are requested, and the answers to the questions.

Experiments

a) TCP throughput:

For the measurements in this experiment place your mobile (or second laptop/desktop) close to the AP and run the application iperf3 with the option “-s” (server mode).

Select five locations in your home with a different distance from the AP of your wireless network. The locations should include one close to the AP and one far from the AP, while the other three locations are in intermediate distances. Set the laptop in each of the locations and measure 1) the signal strength from your AP, 2) the Wi-Fi link speed, and the TCP throughput. The signal strength can be measured using the Wi-Fi analyzer application. The speed of your Wi-Fi interface can be found from the Wi-Fi status on your laptop/desktop computer. The TCP throughput can be measured using the iperf3 application running on your laptop with the option “-c” (client mode), giving the IP address of the mobile running the iperf server and the server’s port number. For the throughput, take the average value from at least 5 measurements, with each measurement having a duration of 10 seconds.

Include in a table for each location where you conducted the measurements the distance (approximate) from the AP and the three measurements: signal strength, Wi-Fi link speed, and average TCP throughput.

Create two graphs: The first graph showing the signal strength and the W-Fi link speed (in the same graph) as a function of the distance from the AP and the second graph showing the Wi-Fi link speed and the average throughput as a function of the signal strength.

Answer the following questions:

i) What is the shape of the two graphs? Did you expect this shape?
ii) How would the first graph change if the horizontal axis showed the logarithm of the distance?
iii) In the second graph (B), how does the throughput compare with the Wi-Fi link speed? Did you expect this relation?

Notes:

- The default transport protocol for iperf3 is TCP. This is the protocol that you will use for the measurements in this experiment.
- Perform the throughput measurements using the iperf3 options "-t XX" and "-i YY". For example, the options "-t 60 -i 10" perform 6 measurements each with duration 10 seconds. See the page https://iperf.fr/iperf-doc.php for a description of the iperf3 options.

b) TCP throughput in both directions

Select a location far from the AP where you will place the laptop for the next measurements. Find the Wi-Fi link speed of your laptop and measure the TCP throughput from the laptop to the mobile and in the opposite direction, from the mobile to the laptop. The TCP throughput from the mobile to the laptop can be measured using the iperf3 option "-R" (reverse) with which the server is the sender and the client is the receiver.

c) UDP throughput

The next measurements, at the same location where the measurements in experiment b) were performed, will use the UDP protocol (iperf3 option "-u").

Measure the UDP throughput from the laptop to the mobile for iperf3 sending rate from 10 Mbps up to twice the Wi-Fi link speed, with step S. The iperf3 sending rate (referred to as "bandwidth") can be set using the option "-b [YYY]M", where YYY is the sending rate and M indicates Mbps. The step S should be S = 10 Mbps if the Wi-Fi speed is <= 50 Mbps, S = 20 Mbps if the Wi-Fi speed is >50 Mbps and <= 200 Mbps, and S = 50 Mbps if the speed is >200 Mbps.

Add the above measurements in a table and create a graph of the UDP throughput as a function of the iperf3 sending rate (value YYY in the iperf3 option "-b").

d) UDP throughput – reverse direction

Repeat the measurements of experiment c) but with the server (mobile) being the sender and the client (laptop) being the receiver. Report the receiving rate at the client.

Based on the measurements in b), c), and d) answer the following questions:

i) How does the UDP throughput from the laptop to the mobile compare with the Wi-Fi link speed and with the TCP throughput? Did you expect this relation?

ii) How does the UDP throughput from the mobile to the laptop compare with the UDP throughput from the laptop to the mobile?

iii) For the different values of the iperf3 UDP sending rate, for both directions, what is the bottleneck (if one exists) between the mobile and the laptop?